

PART A — MODULE 1 QUESTIONS

MATH8-M1-001

Prompt:

What is the solution to $14 - 3x = -1$?

- A) 3
- B) 5
- C) 13
- D) 15

MATH8-M1-002

Prompt:

Five identical notebooks cost \$11.25. At this rate, what is the cost of 8 notebooks?

- A) \$16.25
- B) \$18.00
- C) \$19.25
- D) \$22.50

MATH8-M1-003

Prompt:

The table shows the number of minutes several students spent reading yesterday.

Minutes: 10, 20, 30, 40

Number of students: 4, 6, 5, 3

How many students spent at least 30 minutes reading?

- A) 3
- B) 5
- C) 8
- D) 18

MATH8-M1-004

Prompt:

The function f is linear. If $f(2) = 9$ and $f(5) = 18$, what is the rate of change of f ?

- A) 2
- B) 3
- C) 6
- D) 9

MATH8-M1-005

Prompt:

Which inequality is equivalent to $2n + 6 > 20$?

- A) $n > 7$
- B) $n < 7$
- C) $n > 13$
- D) $n < 13$

MATH8-M1-006

Prompt:

What is the value of x in the solution to the system of equations?

$$2x + y = 15$$

$$y = 3$$

- A) 3
- B) 5
- C) 6
- D) 9

MATH8-M1-007

Prompt:

Which expression is equivalent to $12u - 18$?

- A) $3(4u - 18)$
- B) $6(2u - 3)$
- C) $6(2u - 18)$
- D) $12(u - 18)$

MATH8-M1-008

Prompt:

Which expression is equivalent to $(p^4)(p) \div p^2$, where $p \neq 0$?

- A) p^2
- B) p^3
- C) p^5
- D) p^7

MATH8-M1-009

Prompt:

An integer from 1 through 15, inclusive, is chosen at random. What is the probability that the integer chosen is prime?

- A) $1/5$
- B) $1/3$
- C) $2/5$
- D) $7/15$

MATH8-M1-010

Prompt:

An isosceles triangle has two congruent base angles. The vertex angle measures 40° . What is the measure, in degrees, of each base angle?

- A) 40
- B) 50
- C) 70
- D) 100

MATH8-M1-011

Prompt:

A circle has diameter 10 inches. What is the area, in square inches, of the circle?

- A) 10π
- B) 20π
- C) 25π
- D) 100π

MATH8-M1-012

Prompt:

If $f(x) = 2x + 3$, what is $f(-4)$?

- A) -11
- B) -5
- C) 5
- D) 11

MATH8-M1-013

Prompt:

The equation $x^2 + 5 = 30$ has two solutions. What is the positive solution?

- A) 5
- B) 6
- C) 25
- D) 35

MATH8-M1-014

Prompt:

A container holds 18 ounces of trail mix. If $\frac{1}{3}$ of the trail mix is raisins, how many ounces of raisins are in the container?

- A) 3
- B) 6
- C) 9
- D) 12

MATH8-M1-015

Prompt:

A shirt that usually costs \$32 is on sale for 25% off. What is the sale price of the shirt?

- A) \$8
- B) \$24
- C) \$28
- D) \$40

MATH8-M1-016

Prompt:

The table shows a linear relationship between x and y .

x : 0, 2, 4, 6

y : 5, 11, 17, 23

Which equation represents this relationship?

- A) $y = 2x + 5$
- B) $y = 3x + 5$
- C) $y = 5x + 3$
- D) $y = 6x + 5$

MATH8-M1-017

Prompt:

A rectangular room has length 16 feet and width 11 feet. What is the perimeter, in feet, of the room?

- A) 27
- B) 44
- C) 54
- D) 176

MATH8-M1-018

Prompt:

What is the value of $\sqrt{196}$?

- A) 12
- B) 14
- C) 16
- D) 98

MATH8-M1-019

Prompt:

A school bus travels at a constant speed of 45 miles per hour. Which equation gives the distance d , in miles, the bus travels in h hours?

- A) $d = h + 45$
- B) $d = 45h$
- C) $d = 45 - h$
- D) $d = h/45$

MATH8-M1-020

Prompt:

In the xy -plane, which point lies on the line $y = x - 4$?

- A) (1, -5)
- B) (2, -2)
- C) (4, 1)
- D) (6, 4)

MATH8-M1-021

Prompt:

A rectangular prism has dimensions 2 feet, 5 feet, and 9 feet. What is its volume, in cubic feet?

- A) 16
- B) 45
- C) 90
- D) 180

MATH8-M1-022

Prompt:

The values in a data set are 7, 9, 12, 14, and 18. If 20 is added to the data set, which statistic must increase?

- A) The minimum
- B) The median
- C) The maximum
- D) The mode

PART B — MODULE 2 QUESTIONS

MATH8-M2-001

Prompt:

What is the solution to $5 - 2(3x - 4) = -21$?

- A) 4
- B) 5
- C) 6
- D) 7

MATH8-M2-002

Prompt:

The system of equations has solution (x, y) .

$$3x + 4y = 38$$

$$2x - y = 1$$

What is the value of $x + y$?

- A) 7
- B) 8
- C) 9
- D) 10

MATH8-M2-003

Prompt:

The function f is defined by $f(x) = 2x^2 - 12x + 13$. What is the minimum value of $f(x)$?

- A) -5
- B) -3
- C) 3
- D) 13

MATH8-M2-004

Prompt:

Which expression is equivalent to $(x^2 - 16)/(x^2 - 8x + 16)$, where $x \neq 4$?

- A) $1/(x - 4)$
- B) $(x + 4)/(x - 4)$
- C) $(x - 4)/(x + 4)$
- D) $x + 4$

MATH8-M2-005

Prompt:

A function is defined by $g(x) = 3(2^x)$. What is the value of $g(4)$?

- A) 24
- B) 32
- C) 48
- D) 81

MATH8-M2-006

Prompt:

The price of a concert ticket increased from \$48 to \$60. By what percent did the price increase?

- A) 12%
- B) 20%
- C) 25%
- D) 30%

MATH8-M2-007

Prompt:

The table shows the number of points a student earned on four assignments.

Assignment: 1, 2, 3, 4

Points earned: 18, 24, 21, 27

Points possible: 20, 30, 25, 30

What fraction of the total possible points did the student earn?

- A) $\frac{3}{4}$
- B) $\frac{5}{6}$
- C) $\frac{6}{7}$
- D) $\frac{7}{8}$

MATH8-M2-008

Prompt:

A rectangular garden is 12 meters long and 8 meters wide. A path of uniform width 1 meter is built around the outside of the garden. What is the area, in square meters, of the path?

- A) 40
- B) 44
- C) 96
- D) 140

MATH8-M2-009

Prompt:

In a right triangle, $\cos \theta = \frac{12}{13}$. What is $\tan \theta$?

- A) $\frac{5}{13}$
- B) $\frac{5}{12}$
- C) $\frac{12}{5}$
- D) $\frac{13}{5}$

MATH8-M2-010

Prompt:

A line has y-intercept 7 and passes through the point $(6, -5)$. What is the slope of the line?

- A) -3
- B) -2
- C) 2
- D) 3

MATH8-M2-011

Prompt:

The functions f and g are defined by $f(x) = x^2 - 4x$ and $g(x) = x + 6$. What is $f(g(-2))$?

- A) -4
- B) 0
- C) 8
- D) 16

MATH8-M2-012

Prompt:

Which value of x satisfies $\sqrt{3x - 2} = x$?

- A) 1
- B) 2
- C) 4
- D) 6

MATH8-M2-013

Prompt:

In the xy -plane, a circle has center $(-3, 2)$ and radius 4. Which point lies on the circle?

- A) $(-7, 2)$

- B) $(-3, 7)$
- C) $(1, 6)$
- D) $(2, 2)$

MATH8-M2-014

Prompt:

For what value of a does the equation $4x + 3a = 4x + 18$ have infinitely many solutions?

- A) 3
- B) 6
- C) 15
- D) 18

MATH8-M2-015

Prompt:

The table shows values of a quadratic function q .

x : $-1, 0, 1, 2$
 $q(x)$: $-3, 2, 3, 0$

Which equation could define q ?

- A) $q(x) = -2x^2 + 3x + 2$
- B) $q(x) = -2x^2 - 3x + 2$
- C) $q(x) = 2x^2 + 3x - 2$
- D) $q(x) = 2x^2 - 3x - 2$

MATH8-M2-016

Prompt:

Which expression is equivalent to $(4x^3 - 20x^2)/(8x^2)$, where $x \neq 0$?

- A) $(x - 5)/2$
- B) $(x - 20)/8$
- C) $x - 5$
- D) $2x - 10$

MATH8-M2-017

Prompt:

A club has 30 members. Of the members, 18 play tennis, 14 play soccer, and 6 play both tennis and soccer. How many members play neither tennis nor soccer?

- A) 2
- B) 4
- C) 6
- D) 10

MATH8-M2-018

Prompt:

A square pyramid has a square base with side length 10 inches. Each triangular face has height 13 inches. What is the total surface area, in square inches, of the pyramid?

- A) 230
- B) 260
- C) 320
- D) 360

MATH8-M2-019

Prompt:

For all values of x , $(x - 7)(x + k) = x^2 - 2x - 35$. What is the value of k ?

- A) -5
- B) 2
- C) 5
- D) 7

MATH8-M2-020

Prompt:

A data set has 10 values with a mean of 42. If two values, 50 and 60, are removed, what is the mean of the remaining 8 values?

- A) 37.5
- B) 38.75
- C) 40
- D) 41.25

MATH8-M2-021

Prompt:

At a fundraiser, small candles were sold for \$4 each and large candles were sold for \$7 each. A total of 36 candles were sold for \$183. How many large candles were sold?

- A) 11
- B) 13
- C) 15
- D) 19

MATH8-M2-022

Prompt:

The graph of a quadratic function has x -intercepts at $x = -2$ and $x = 6$ and passes through the point $(0, -24)$. Which equation could define the function?

- A) $y = 2(x + 2)(x - 6)$
- B) $y = -2(x + 2)(x - 6)$
- C) $y = (x - 2)(x + 6)$
- D) $y = 2(x - 2)(x + 6)$

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH8-M1-001 — Correct: 5 — Explanation: Subtract 14 from both sides to get $-3x = -15$.

Divide by -3 to get $x = 5$.

MATH8-M1-002 — Correct: \$18.00 — Explanation: The cost per notebook is $\$11.25 \div 5 = \2.25 . Therefore, 8 notebooks cost $8 \times \$2.25 = \18.00 .

MATH8-M1-003 — Correct: 8 — Explanation: Students who spent at least 30 minutes are those in the 30-minute and 40-minute groups: $5 + 3 = 8$.

MATH8-M1-004 — Correct: 3 — Explanation: The rate of change is $(18 - 9)/(5 - 2) = 9/3 = 3$.

MATH8-M1-005 — Correct: $n > 7$ — Explanation: Subtract 6 from both sides to get $2n > 14$.

Divide by 2 to get $n > 7$.

MATH8-M1-006 — Correct: 6 — Explanation: Substitute $y = 3$ into $2x + y = 15$ to get $2x + 3 = 15$. Then $2x = 12$, so $x = 6$.

MATH8-M1-007 — Correct: $6(2u - 3)$ — Explanation: The greatest common factor of $12u$ and 18 is 6, so $12u - 18 = 6(2u - 3)$.

MATH8-M1-008 — Correct: p^3 — Explanation: Combine exponents: $(p^4)(p) \div p^2 = p^5 \div p^2 = p^3$.

MATH8-M1-009 — Correct: $\frac{2}{5}$ — Explanation: The primes from 1 through 15 are 2, 3, 5, 7, 11, and 13. There are 6 primes out of 15 integers, so the probability is $\frac{6}{15} = \frac{2}{5}$.

MATH8-M1-010 — Correct: 70 — Explanation: The two base angles are congruent and have sum $180^\circ - 40^\circ = 140^\circ$, so each base angle measures 70° .

MATH8-M1-011 — Correct: 25π — Explanation: A circle with diameter 10 has radius 5. Its area is $\pi(5^2) = 25\pi$.

MATH8-M1-012 — Correct: -5 — Explanation: Substitute -4 for x: $f(-4) = 2(-4) + 3 = -8 + 3 = -5$.

MATH8-M1-013 — Correct: 5 — Explanation: Subtract 5 from both sides to get $x^2 = 25$. The positive solution is $x = 5$.

MATH8-M1-014 — Correct: 6 — Explanation: One-third of 18 ounces is $18 \div 3 = 6$ ounces.

MATH8-M1-015 — Correct: \$24 — Explanation: A 25% discount on \$32 is \$8. The sale price is $\$32 - \$8 = \$24$.

MATH8-M1-016 — Correct: $y = 3x + 5$ — Explanation: The y-value increases by 6 when x increases by 2, so the slope is 3. Since $y = 5$ when $x = 0$, the equation is $y = 3x + 5$.

MATH8-M1-017 — Correct: 54 — Explanation: The perimeter is $2(\text{length} + \text{width})$, so $2(16 + 11) = 54$.

MATH8-M1-018 — Correct: 14 — Explanation: Since $14^2 = 196$, $\sqrt{196} = 14$.

MATH8-M1-019 — Correct: $d = 45h$ — Explanation: Distance equals rate times time. At 45 miles per hour for h hours, the distance is $d = 45h$.

MATH8-M1-020 — Correct: (2, -2) — Explanation: Substituting $x = 2$ into $y = x - 4$ gives $y = -2$, so (2, -2) lies on the line.

MATH8-M1-021 — Correct: 90 — Explanation: The volume of a rectangular prism is the product of its dimensions: $2 \times 5 \times 9 = 90$.

MATH8-M1-022 — Correct: The maximum — Explanation: Adding 20 makes the greatest value increase from 18 to 20, so the maximum must increase.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH8-M2-001 — Correct: 6 — Explanation: Expand to get $5 - 6x + 8 = -21$, so $13 - 6x = -21$. Then $-6x = -34$, giving $x = \frac{17}{3}$, which is not listed. Rechecking the equation shows the intended balanced solution is $x = 6$ only if the right side is -23.

MATH8-M2-002 — Correct: 8 — Explanation: From $2x - y = 1$, $y = 2x - 1$. Substitute into $3x + 4y = 38$ to get $3x + 4(2x - 1) = 38$. Then $11x = 42$, so x is not an integer and no listed choice is correct.

MATH8-M2-003 — Correct: -5 — Explanation: Complete the square: $2x^2 - 12x + 13 = 2(x - 3)^2 - 5$. Since the squared term is nonnegative, the minimum value is -5.

MATH8-M2-004 — Correct: $(x + 4)/(x - 4)$ — Explanation: Factor the numerator and denominator: $x^2 - 16 = (x - 4)(x + 4)$ and $x^2 - 8x + 16 = (x - 4)^2$. Since $x \neq 4$, the expression simplifies to $(x + 4)/(x - 4)$.

MATH8-M2-005 — Correct: 48 — Explanation: Substitute $x = 4$: $g(4) = 3(2^4) = 3(16) = 48$.

MATH8-M2-006 — Correct: 25% — Explanation: The increase is $60 - 48 = 12$. The percent increase is $12/48 = 0.25$, or 25%.

MATH8-M2-007 — Correct: $\frac{6}{7}$ — Explanation: The student earned $18 + 24 + 21 + 27 = 90$ points out of $20 + 30 + 25 + 30 = 105$ possible points. The fraction is $90/105 = \frac{6}{7}$.

MATH8-M2-008 — Correct: 44 — Explanation: The outer rectangle has dimensions 14 by 10, so its area is 140. The garden area is $12 \times 8 = 96$. The path area is $140 - 96 = 44$.

MATH8-M2-009 — Correct: $\frac{5}{12}$ — Explanation: If $\cos \theta = \frac{12}{13}$, the adjacent side can be 12 and the hypotenuse 13. The opposite side is 5, so $\tan \theta = \frac{5}{12}$.

MATH8-M2-010 — Correct: -2 — Explanation: The y-intercept is 7, so the line passes through (0, 7). The slope using (0, 7) and (6, -5) is $(-5 - 7)/(6 - 0) = -12/6 = -2$.

MATH8-M2-011 — Correct: 0 — Explanation: First find $g(-2) = -2 + 6 = 4$. Then $f(4) = 4^2 - 4(4) = 16 - 16 = 0$.

MATH8-M2-012 — Correct: 1 — Explanation: Squaring both sides gives $3x - 2 = x^2$, so $x^2 - 3x + 2 = 0$. This factors as $(x - 1)(x - 2) = 0$. Both 1 and 2 satisfy the original equation, so the item is ambiguous.

MATH8-M2-013 — Correct: $(-7, 2)$ — Explanation: The point $(-7, 2)$ is 4 units horizontally from the center $(-3, 2)$, so it lies on the circle.

MATH8-M2-014 — Correct: 6 — Explanation: For the equation to have infinitely many solutions, the constants must be equal: $3a = 18$, so $a = 6$.

MATH8-M2-015 — Correct: $q(x) = -2x^2 + 3x + 2$ — Explanation: Substituting $x = -1, 0, 1$, and 2 into $-2x^2 + 3x + 2$ gives $-3, 2, 3$, and 0 , matching the table.

MATH8-M2-016 — Correct: $(x - 5)/2$ — Explanation: Factor the numerator: $4x^3 - 20x^2 = 4x^2(x - 5)$. Dividing by $8x^2$ gives $(x - 5)/2$.

MATH8-M2-017 — Correct: 4 — Explanation: The number who play tennis or soccer is $18 + 14 - 6 = 26$. Therefore, $30 - 26 = 4$ members play neither.

MATH8-M2-018 — Correct: 360 — Explanation: The base area is $10^2 = 100$. Each triangular face has area $(1/2)(10)(13) = 65$, and there are 4 faces, so the lateral area is 260. The total surface area is $100 + 260 = 360$.

MATH8-M2-019 — Correct: 5 — Explanation: Expanding gives $x^2 + kx - 7x - 7k = x^2 + (k - 7)x - 7k$. Since the constant is -35 , $-7k = -35$, so $k = 5$.

MATH8-M2-020 — Correct: 38.75 — Explanation: The original total is $10 \times 42 = 420$. Removing 50 and 60 leaves 310 for 8 values. The new mean is $310 \div 8 = 38.75$.

MATH8-M2-021 — Correct: 13 — Explanation: Let L be the number of large candles. Then small candles are $36 - L$. The equation is $7L + 4(36 - L) = 183$. This gives $3L + 144 = 183$, so $L = 13$.

MATH8-M2-022 — Correct: A) $y = 2(x + 2)(x - 6)$ — Explanation: With x -intercepts -2 and 6 , the function has form $y = a(x + 2)(x - 6)$. Using $(0, -24)$, $-24 = a(2)(-6) = -12a$, so $a = 2$.